



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 413 : Computer Graphics

Batch: 21st

Date: 14/01/2025

Midterm Examination

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

1. (a) Define the followings:

3 × 1 = 3

- i. Refresh Rate
- ii. Resolution
- iii. Aspect Ratio

✓ (b) Differentiate between Raster scan and Random scan display devices.

3

7. (a) What are the disadvantages of DDA algorithm?

2

✓ (b) Find out the points needed to generate a line drawn from (2, 3) to (6, 5) by using DDA algorithm.

4

3. (a) Explain the 8 ways symmetric properties of a circle.

3

(b) Write the steps for Bresenham's line drawing algorithm.

3

4. (a) State the equation for rotation of a point (x, y) about an arbitrary pivot position (x_r, y_r).

2

✓ (b) Given the center point coordinates (0, 0) and radius 8, generate all the points to form a circle using the Mid Point circle drawing algorithm.

4

5. (a) Define Geometric Transformation.

2

✓ (b) Derive the transformation factor for the following transformation:

4

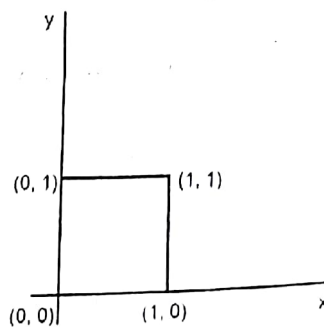


Fig. 1

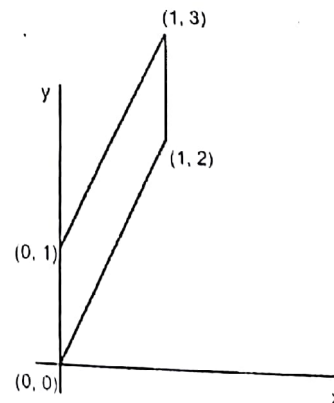


Fig. 2

6. (a) Define 2D Rotation. State the transformation matrix to reflect a point (x, y) about the x and y axis. 2
- (b) Scale the line with endpoints $P_1(2, 3)$ and $P_2(4, 6)$ by factors $S_x = 2$ and $S_y = 2$ relative to the fixed point $F(3, 4)$ and find the new endpoints. 4
7. (a) Prove that successive scaling operations are multiplicative. 3
- (b) Explain the general two-dimensional pivot-point rotation with necessary figures. 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 415: Communication Engineering

Batch : 21st

Date: 04/01/2025

Mid-Term Examination

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks : 30

Answer any five questions:

5 × 6 = 30

- ✓ 1. (a) Write the types of error with example. 3
(b) Draw a table for Simple Parity-Check Code where K = 4 and N = 5. 3 6
2. (a) Illustrate the Division in Cyclic Redundancy Check (CRC) Encoder. 3
(b) State the algorithm to calculate the traditional checksum. 3
- ✓ 3. ✓ (a) Define Connectionless and Connection-Oriented Protocol. 3 2
(b) Draw and explain FSM for the Stop-and-Wait Protocol. 3
- ✓ 4. (a) Explain Control field format for different HDLC frame types. 3
(b) Compare and contrast byte-stuffing and bit-stuffing. 3 3
5. (a) Why the throughput of Slotted ALOHA is 2 times greater than Pure ALOHA? Explain. 3
(b) Sketch the flow diagram for three persistence methods. 3
- ✓ 6. (a) Write the importance of NAV in CSMA/CA. 3
(b) Why collision is an issue in Random access protocol but not in Channelization protocol? 3 2
- ✓ 7. (a) Draw and explain Ethernet frame. 3
(b) What are the common Standard Ethernet implementations? 3



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 21st

Date: 01/01/2025

Mid Term Examination

Semester: Fall 2024

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following:

5x6= 30

- 1 (a) Explain the significance of AI in the context of computer vision. 2
(b) How is AI applied to enhance both machine translation and logical reasoning? 4

- 2 (a) List the main differences between knowledge-based system and conventional programming. 3
(b) Perform k-means clustering for the following dataset: 3

Feature_1	Feature_2	Feature_3
1.0	2.0	3.0
5.0	8.0	7.0
1.5	2.5	3.5
6.0	7.0	8.0
0.5	1.5	2.5

- 3 (a) Determine whether the following sentence, S3: $(P \ \& \ Q) \rightarrow R \vee \sim Q$ is (a) satisfiable, (b) contradictory, or (c) valid. 3
(b) State the properties of statements in knowledge representation. 3

- 4 (a) Draw the complete structure of a rule-based expert system. 2
(b) Write the predicate logic for the following statements: 4x1= 4
i. No student failed the exam.
ii. There exists a city where all the houses are green.
iii. For every book, there is an author who wrote it.
iv. There exists a restaurant where all the dishes are vegetarian.

- 5 (a) Outline the problems with semantic nets. 2
(b) Demonstrate the match-fire cycle using backward chaining for the following database and knowledge base to prove whether C is true. 4

Data Base (DB):

D, G, H

Knowledge Base (KB):

$A \ \& \ B \rightarrow C$

$D \rightarrow A$

$E \ \& \ F \rightarrow B$

$G \rightarrow E$

$H \rightarrow F$

- 6 (a) How would you describe AI in terms of human-like intelligence? 2
(b) Create a frame representing the information: "Mary is a doctor, she specializes in pediatrics, she lives in a suburban house, and she loves to 4

read mystery novels."

- x 7 (a) State the advantages and disadvantages of frames in knowledge representation. 3
- (b) Explain the methods used to resolve conflict resolution in expert system. 3



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE471: Computer and Network Security

Batch: 21st

Date: 07/01/2025

Mid-Term Examination

Semester: Fall 2024

Time: 1 Hour 30 minutes

Total Marks: 30

Answer any five questions of the following.

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5912

5 × 6 = 30

1. a) State the differences between active and passive attack. 2
b) Suppose Alice and Bob want to use symmetric key encryption for secure communication. They want to establish their secret key using Diffie-Hellman Key Exchange Algorithm. Show the complete key establishment process if $g = 11$, $n = 13$, $x = 5$ and $y = 2$. 4
2. a) What are the limitations of classical encryption algorithms? 2
b) Discuss the key management problem in symmetric key encryption. 2
c) State the differences between cryptanalysis and decryption. 2
3. a) Let $p = 7$, $q = 13$ and $e = 5$. Calculate the private and public key using RSA algorithm. 2
b) Suppose message $M = 10$, show the complete encryption and decryption process using the keys you have just calculated. 4
4. a) What is passive attack? 1
b) How can an attacker perform a man-in-the-middle (MITM) attack in an unencrypted communication environment with ensuring the attack remains completely unnoticeable by the authorized users? Explain the process in detail. 2
c) The bold text is encrypted using Shift Cipher. Show the cryptanalysis and determine the plaintext. Also determine the key that was used for encryption. 'SJB DJFW SJB RJ' 3
5. a) State the six key characteristics of Hashing. 2
b) Explain how Alice can use digital signature to sign her message $m = 'hi'$ and encrypt it before sending it to Bob. Also, demonstrate how Bob will verify Alice's message and perform decryption. Assume that both Alice and Bob are aware of each other's public keys. 4
6. a) What is brute force attack? 1
b) Explain the CIA triad. 1
c) Show the decryption process of the following text using Rail Fence Cipher where depth = 2. You need to use all the related formulas. 2
EETOANTCLCRMGEI 3
7. a) What is rubber ducky? Explain how social engineering is important to perform attack using rubber ducky. 2
b) Show the encryption process of the following message and determine the ciphertext 'C' using Row Column Cipher where row = 4, column = 4 and key = 3124. 2
UNCHARACTERISTIC
c) Use the same information and show the complete decryption process of the ciphertext 'C'. 2

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 417: Compiler Design

Batch: 21st

Date: 12/01/2025

Mid-Term Examination

Semester: Fall 2024

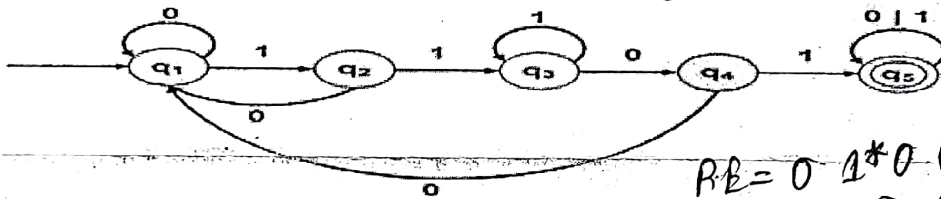
Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions.

Marks: 5 x 6 = 30

1. (a) Demonstrate the syntax analysis of a compiler with an example. 3
 (b) Draw the epsilon-NFA for the following Regular Expression (RE): 3
 $((a^*b^*)^* + (a^* + b^*)^*)^*$
2. (a) State the goal of a scanner of a Compiler. 3
 (b) Briefly explain how a DFA process strings. 3
3. (a) Distinguish between DFA and NFA. 3
 (b) Briefly explain how an automaton recognizes a set of keywords. 3
4. Find the RE for the following Automata by eliminating states: 6



$$RE = 0^* 1^* 0^* (1^* + 0^*)^*$$

$$= (1^* 0^*)^* (1^* + 0^*)^*$$

5. Convert the following epsilon-NFA to a DFA: 6

States	ϵ	a	b	c
$\rightarrow p$	{q, r}	{}	{q}	{r}
q	{r}	{p}	{r}	{q}
r*	{}	{s}	{p}	{r}
s*	{p}	{s}	{}	{r}

6. Derive the DFA with transition table for the following expression: 3x2 = 6
 - (i) Start with 1011001
 - (ii) End with 1011001
 - (iii) 1011001 be a substring of any string

7. Convert the following NFA to a DFA: 6

States	Inputs	
	0	1
$\rightarrow p \cdot$	{p, q}	{p, r}
q	{r, s}	{s}
$\cdot r^* \cdot$	{r}	{r}
s	{}	{t}
$\cdot t^* \cdot$	{u}	{t}
u	{r}	{s}